

Synopsis - Week 4: Basic Logic Gates and Finite State Machines (FSMs)

Prof. Jordan C. Hanson

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1 Introductions to Connect and Disconnect to PYNQ-Z1 System-on-a-Chip (SoC)

1. Connect to the PYNQ-Z1 board in the usual fashion, *ensuring that the system is logged out* before starting a new kernel.
2. If troubleshooting is needed, consult lab partners or the professor.

2 Complex Boolean Logic Functions

1. Close the FSM generator notebook tab, and shut it down using the **Running** tab in the PYNQ-Z1 homescreen.
2. Start a new Jupyter notebook (New → Python3), create a python dictionary entitled *function* and use the boolean generator to perform multiple logic operations:

```
from pynq.overlays.logictools import LogicToolsOverlay
logictools_olay = LogicToolsOverlay('logictools.bit')
function = {'f1': 'LD0 = ~( ~(PB0 & ~(PB0 & PB1)) & ~(PB1 & ~(PB0 & PB1)) )'}
boolean_generator = logictools_olay.boolean_generator
boolean_generator.setup(function)
boolean_generator.run()
```

In a separate cell, you can add the stop function for the boolean generator:

```
boolean_generator.stop()
```

3. Before recording the truth table, draw a diagram below using NAND gates to describe what effect the firmware *should* produce.
4. Now record the truth table, and identify the logic gate by the truth table.
5. Does the logic gate identified match the diagram deduced from reading the code? How can you show that they are equivalent? Construct the proof below.
6. Reset the kernel, and repeat steps 2-5 for the following boolean function:

```
function = {'f1': 'LD0 = ~( ~(PB0 & PB1) & ~(PB0 & PB1) )'}
```

3 Logging Out and Shutting Down

1. Close all Jupyter notebooks, and navigate to the homescreen of the PYNQ-Z1 board.
2. Click New → Terminal.
3. When the terminal pops up, enter **shutdown now** , and wait for the system to disconnect from the laptop. Then close all browser tabs.